Data and Artificial Intelligence
Cyber Shujaa Program

Week 7 Assignment
Regression Models

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Introduction

This week's assignment involved implementing Linear Regression Models using Python in Google Colab. The goal was to understand how regression algorithms can predict numerical values such as house prices based on given features.

The assignment required working with three datasets:

- `homeprices.csv` – used for simple linear regression.
- `areas.csv` – used for generating predictions from the trained model.
- `homeprices-m.csv` – used for multivariate linear regression.

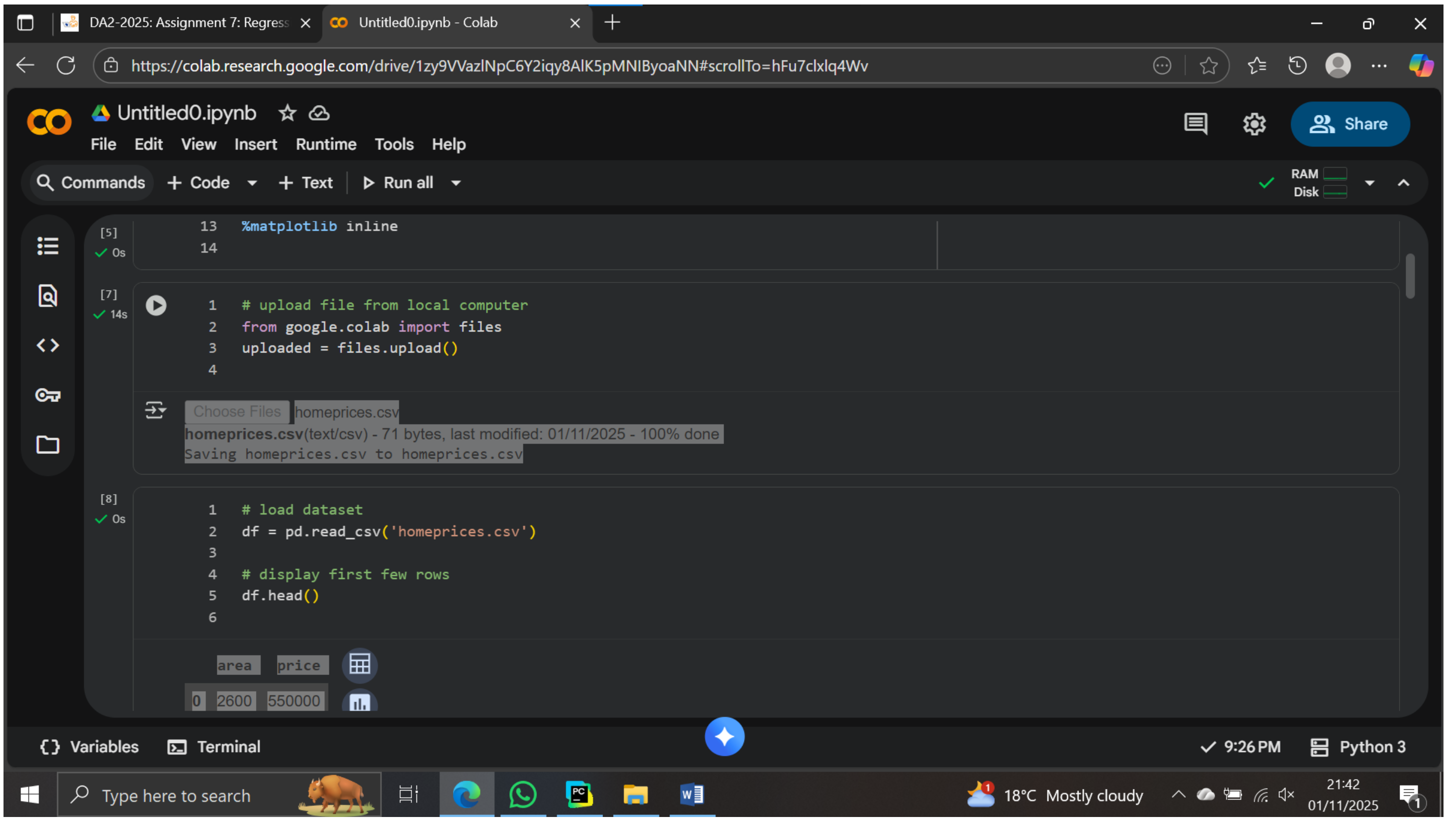
The following sections explain each step of the process and show screenshots of the corresponding code and outputs.

Steps and Implementation

2.1 Loading Libraries and Dataset

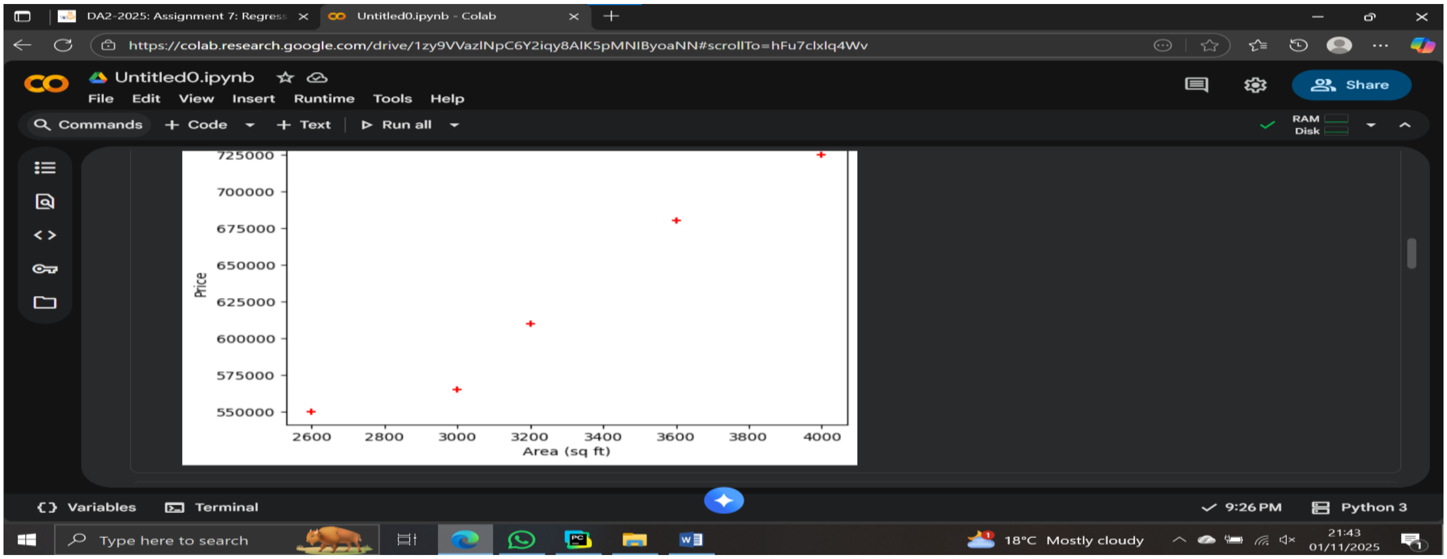
The first step involved importing all necessary Python libraries such as `pandas`, `numpy`, `matplotlib`, and `scikit-learn`.
These libraries were used for data handling, visualization, model building, and evaluation.

The dataset `homeprices.csv` was uploaded to Google Colab using the `files.upload()` method and loaded into a DataFrame.



2.2 Data Visualization

A scatter plot was created to visualize the relationship between area and price.
The plot showed a positive correlation, indicating that larger houses tend to have higher prices.

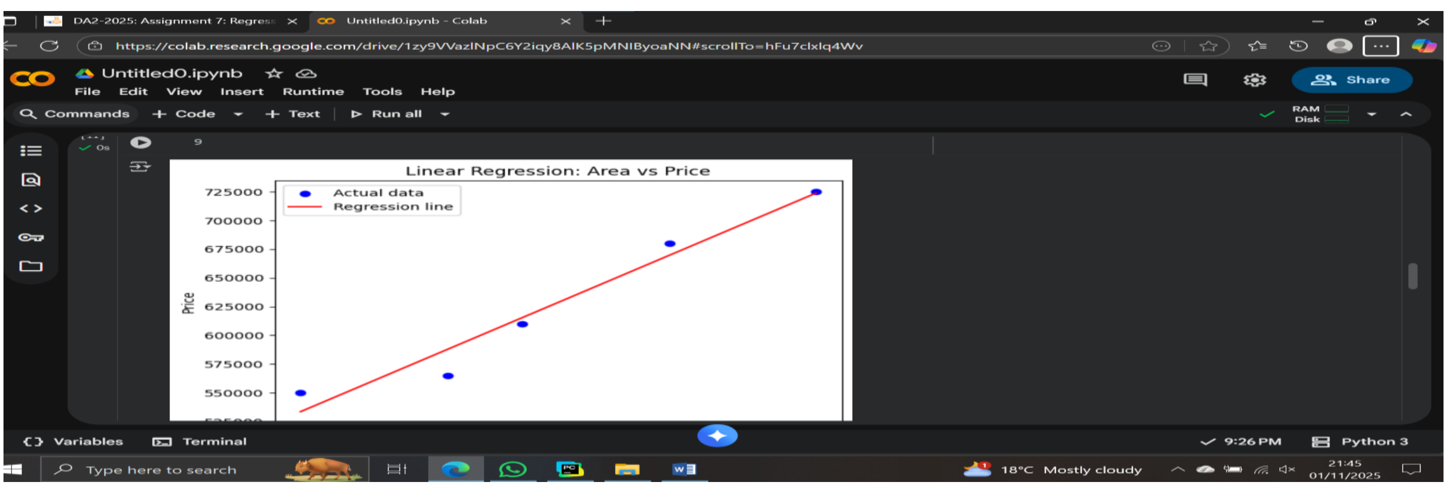


This visual check confirmed that linear regression was suitable for modeling this data since the points roughly followed a straight-line pattern.

2.3 Building the Linear Regression Model

A Linear Regression model was created using Scikit-learn's `LinearRegression()` class.
The model was trained using the area as the independent variable and price as the dependent variable.

After training, the regression line was plotted alongside the actual data points to visualize how well the model fit the data.

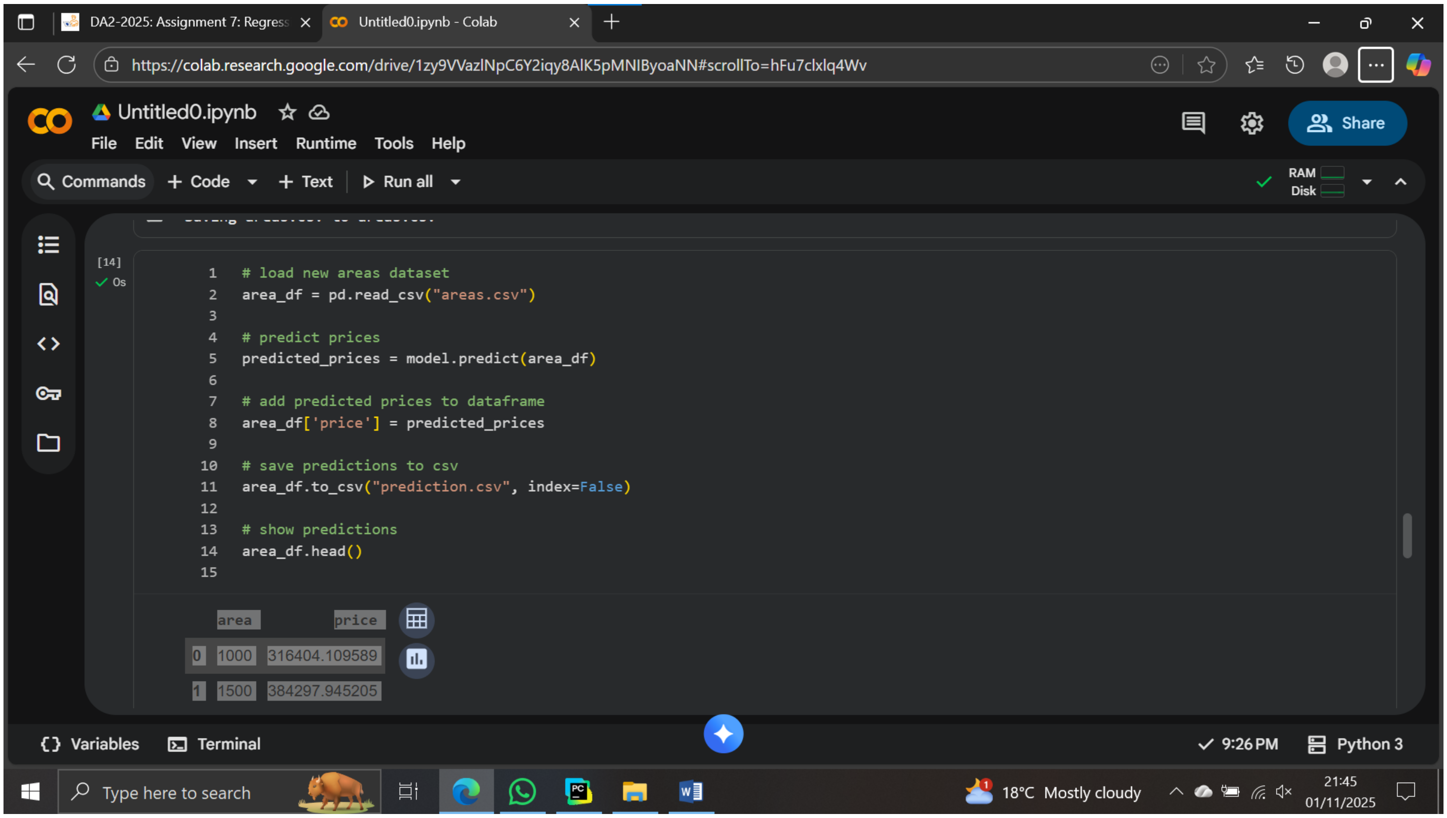


This step demonstrated the relationship between house size and price, where the red line represented the predicted trend and blue dots represented the actual data.

2.4 Predicting Prices for New Areas

The second dataset, `areas.csv`, was uploaded to predict house prices for given areas not included in the training data.
Using the trained regression model, the predicted prices were generated and added to the dataset as a new column.

The final predictions were saved in a new file named `prediction.csv` for review.



This step showed how regression models can be used for real-world predictions once trained.

2.5 Multivariate Regression Model

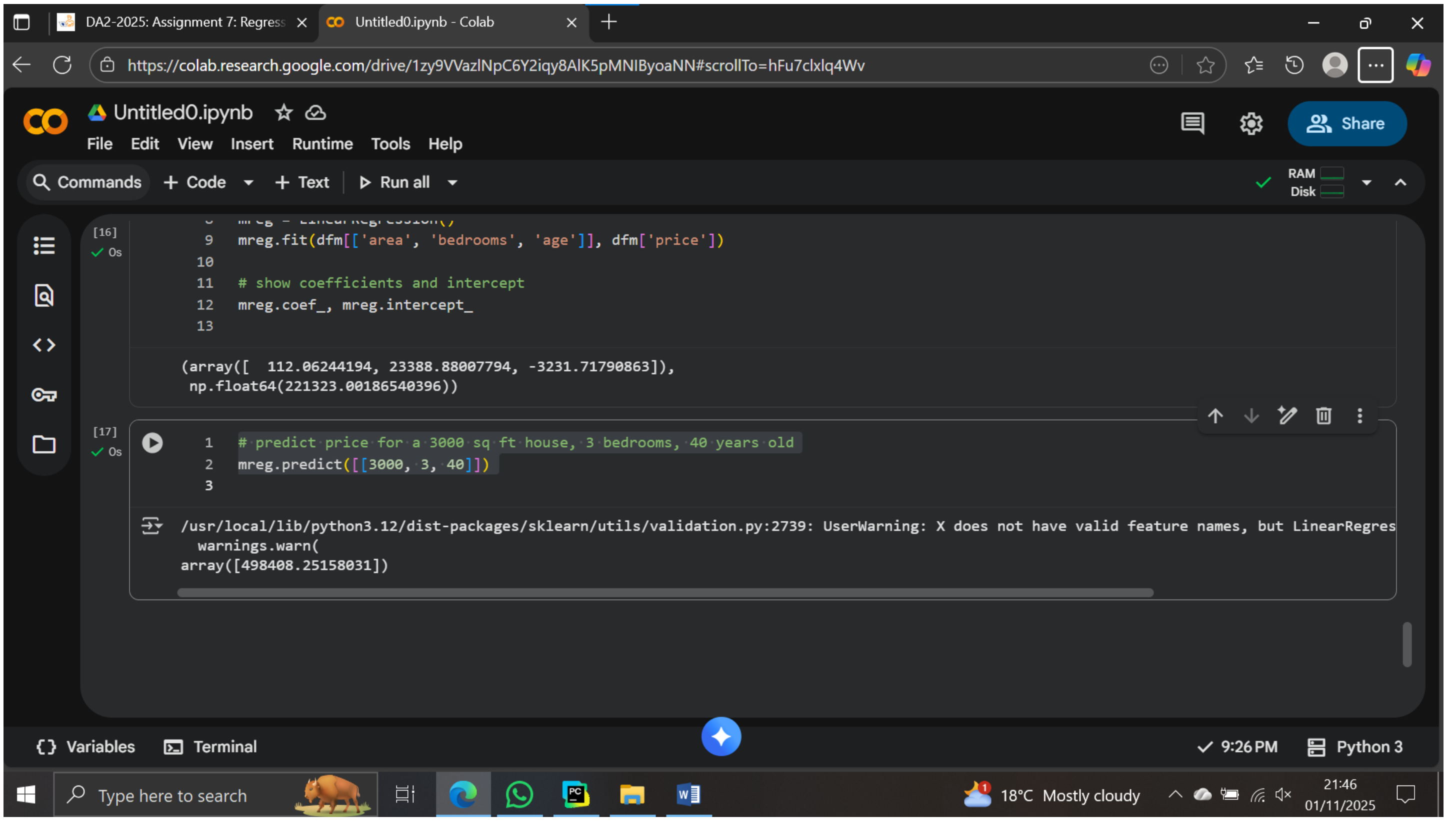
In the final part, a multivariate regression model was developed using `homeprices-m.csv`, which contained three features:

- Area (sq ft)
- Bedrooms
- Age of the house

Before training, missing values in the “bedrooms” column were filled with the median value to ensure data quality.

The model was then trained using all three variables to predict house prices.
This approach allowed the model to consider multiple factors influencing price rather than area alone.

Finally, the model was used to predict the price of a 3000 sq ft house, with 3 bedrooms and 40 years old**,** which resulted in a prediction of approximately 498,408.25 units.



Results and Interpretation

1. The simple linear regression model accurately captured the relationship between area and price, producing realistic predictions.
2. The multivariate regression model improved accuracy by including additional features such as bedrooms and house age.
3. The predicted result for a 3000 sq ft, 3-bedroom, 40-year-old house was 498,408.25, confirming that the model could generalize well to unseen data.
4. The data visualization and model evaluation demonstrated that regression models are effective tools for predictive analytics in real estate pricing.

Conclusion

This assignment provided valuable hands-on experience with regression modeling in Python.

I successfully learned how to:

- Load and clean real-world datasets.
- Visualize data relationships.
- Build and train both simple and multivariate regression models.
- Generate predictions and save them in a new dataset.

Through this project, I gained practical understanding of how machine learning models work, especially in predicting continuous variables such as prices.

These concepts are essential in fields like finance, business, and data analytics, where data-driven predictions inform decision-making.

Colab Notebook Link

Google Colab Notebook:
[https://colab.research.google.com/drive/1zy9VVazlNpC6Y2iqy8AlK5pMNIByoaNN?usp=sharing]